

MSBA 265: Business Analytics Topics

Foundational Module 1: Practical Homework Assignment

Interactive EDA, Data Quality Verification, Data Dictionaries, & Outlier Pipelines

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Course: MSBA 265

Term: Fall 2026

Part 1: Workspace Setup, Git Ingestion, & Procurement

A reproducible analytics workspace was created in Visual Studio Code using Python, a virtual environment, Git, and GitHub. The repository includes a .gitignore, README.md, requirements.txt, data ingestion script, Jupyter Notebook, production cleaning script, exported data dictionary, generated figures, cleaned dataset, and final report.

The French Motor Claims frequency dataset (freMTPL2freq.csv) was programmatically obtained and stored as data/raw_business_data.csv. The downloaded dataset contains **678,013 observations and 12 variables**.

The project was organized so that another user can clone the public GitHub repository, install the required Python dependencies, reproduce the analysis, execute the production cleaning pipeline, and access the generated artifacts.

Part 2: Raw Boundary Audits & Business Dictionary Artifact

2.1 Structural and Data Quality Audit

The raw dataset was examined using df.info() and df.describe().T to evaluate its structure, data types, descriptive statistics, and potential data-quality issues.

The dataset contains 678,013 rows and 12 columns. The missing-value audit identified 0 missing values, and the duplicate audit identified 0 duplicate rows.

The dataset contains the following variables: IDpol, ClaimNb, Exposure, Area, VehPower, VehAge, DrivAge, BonusMalus, VehBrand, VehGas, Density, and Region.

A business data dictionary was created and exported to: reports/data_dictionary.csv

The data dictionary documents each variable's data type, missing-value count, number of unique values, business description, and analytical role.

2.2 Raw Boundary Audit

The raw boundary audit focused on DrivAge, Exposure, and BonusMalus.

- **DrivAge:** Driver age values range from **18 to 100 years**. No negative ages or sentinel null codes were identified, indicating that the observed values fall within the expected raw boundaries.
- **Exposure:** Exposure ranges from approximately **0.002 to 2.01**. No zero or negative Exposure values were identified. Because Exposure represents policy duration, maintaining positive values is essential for rate-based modeling.
- **BonusMalus:** BonusMalus ranges from **50 to 230**. No values below 50 were identified. Higher values may represent legitimate high-risk policyholders rather than data-entry errors.

2.3 Business Interpretation: Exposure, Density, and Area

Exposure represents the amount of time that each insurance policy is active. Policies with longer exposure periods naturally have more opportunity to generate claims. Therefore, Exposure should be incorporated into claim-frequency modeling as a **rate offset** rather than treated as an ordinary predictor.

Density represents the continuous population density of the geographic area associated with a policy, while Area provides a categorical representation of geographic density. Because these two variables encode closely related geographic information, including both in the same linear regression model may introduce redundant information and multicollinearity. Therefore, Density and Area should not both be included in the same linear regression specification because they represent overlapping geographic information.

Part 3: Overriding Automated Rules & Correlation Visuals

3.1 Pearson Correlation Heatmap

A Pearson correlation matrix was calculated for the numeric variables and exported as:

reports/figures/correlation_heatmap.png

Figure 1. Pearson Correlation Heatmap



Correlation Heatmap Interpretation

- **Distribution Shape / Relationship Pattern:** Most numeric variables show weak linear relationships, indicating limited pairwise correlation among the numeric features.
- **Key Metrics / Whiskers:** The most notable relationship in the generated heatmap is between DrivAge and BonusMalus, with a correlation of approximately **-0.48**. Most other pairwise correlations are relatively weak.
- **Recommended Analytics Action:** Keep both DrivAge and BonusMalus for further modeling. Their moderate negative relationship should be considered during model diagnostics, but the correlation alone is not strong enough to justify removing either variable.

3.2 Executive Defense: Why ClaimNb Should Not Be Log-Transformed

ClaimNb represents the actual number of claims associated with each insurance policy. Its distribution is highly right-skewed because approximately 95% of policyholders record zero claims, while relatively few policies contain one or more claims.

Although automated univariate diagnostics may recommend a logarithmic transformation because of this strong right-skewness, this recommendation should be overridden.

A standard natural-log transformation is inappropriate for the observed zero values because:

$$\log(0) = -\infty$$

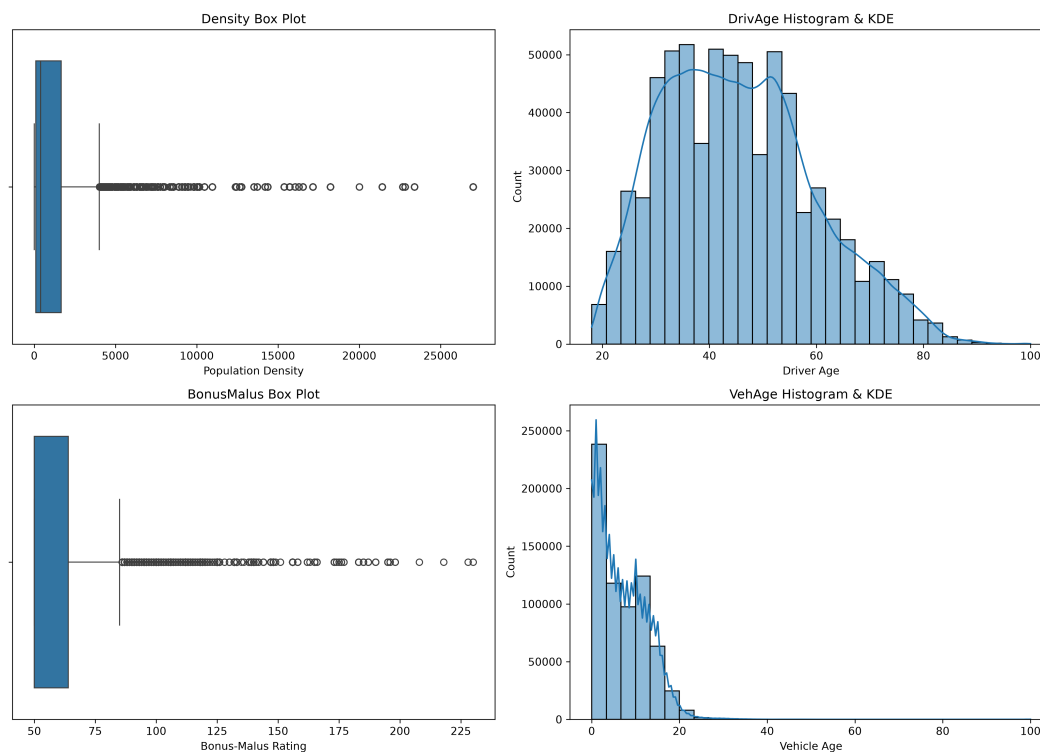
Using a direct logarithmic transformation would therefore make zero observations mathematically undefined. More importantly, ClaimNb is a **discrete count variable**, not a continuous positive-volume measure. Its original values have direct business meaning because they represent the actual number of claim events.

For these reasons, ClaimNb should remain in its original count units rather than being log-transformed. For claim-frequency modeling, ClaimNb is appropriate for a **Poisson rate model using log(Exposure) as an offset**.

Part 4: Distribution Audits & Written Plot Statements

The required distribution plots were generated and exported as:
reports/figures/feature_distributions.png

Figure 2. Required Feature Distribution Audits



The figure contains the Density box plot, DrivAge histogram with KDE, BonusMalus box plot, and VehAge histogram with KDE.

4.1 Density Box Plot

- **Distribution Shape:** Density is extremely right-skewed, with most observations concentrated at relatively low population-density values and a long right tail.
- **Key Metrics / Whiskers:** Based on the dataset analyzed in this project, the Tukey upper fence is approximately **4,007 people/km²**. Observations above this threshold are statistical outliers under the $1.5 \times \text{IQR}$ rule, although high values may still represent valid metropolitan policyholders.
- **Recommended Analytics Action:** Retain valid metropolitan observations during exploratory analysis and consider applying a **$\log(1 + \text{Density})$** transformation before linear regression to reduce the effect of strong right-skewness.

4.2 DrivAge Histogram & KDE

- **Distribution Shape:** DrivAge is approximately symmetric and bell-shaped, with most observations representing adult and middle-aged drivers.
- **Key Metrics / Whiskers:** The median driver age is approximately **44 years**, with **Q1 = 34** and **Q3 = 55**. The observed raw range is **18 to 100 years**.
- **Recommended Analytics Action:** Retain plausible driver-age observations without unnecessary outlier trimming. Apply standard **Z-score scaling** when DrivAge is used in models that benefit from standardized numeric predictors.

4.3 BonusMalus Box Plot

- **Distribution Shape:** BonusMalus is concentrated at lower values and has a pronounced right tail containing higher-risk policyholders.
- **Key Metrics / Whiskers:** The distribution is concentrated near its lower values, while the dataset contains higher BonusMalus observations extending to an observed maximum of **230**.
- **Recommended Analytics Action:** Retain BonusMalus as a continuous risk feature rather than automatically treating high values as data errors. For additional business interpretation, a binary indicator such as **Is_Surcharge = BonusMalus > 100** may also be created.

4.4 VehAge Histogram & KDE

- **Distribution Shape:** VehAge is moderately right-skewed, with most vehicles concentrated among younger ages and a smaller number of older vehicles forming the right tail.
- **Key Metrics / Whiskers:** VehAge has a skewness of approximately **1.15**, with **Q1 = 2 years** and **Q3 = 12 years**. Most observations are concentrated among relatively young vehicles.
- **Recommended Analytics Action:** Retain plausible older vehicles. VehAge may be grouped into interpretable age brackets such as **0–5 years**, **6–10 years**, and **11+ years**, or min-max scaling may be applied when appropriate for modeling.

Part 5: Production Pipeline Scripting & Final PDF Export

5.1 Production Outlier Pipeline

A standalone production-cleaning script was created at: `src/clean_outliers.py`

The script applies the required **Tukey $1.5 \times \text{IQR}$ method** to identify statistical outliers and generates the cleaned dataset: `data/cleaned_business_data.csv`

For a numeric variable, the interquartile range is calculated as: **$\text{IQR} = \text{Q3} - \text{Q1}$**

The Tukey boundaries are then calculated as: **$\text{Lower Bound} = \text{Q1} - 1.5 \times \text{IQR}$**

$\text{Upper Bound} = \text{Q3} + 1.5 \times \text{IQR}$

For the Density variable: **$\text{Q1} = 92$, $\text{Q3} = 1,658$**

Therefore: **$\text{IQR} = 1,658 - 92 = 1,566$**

The resulting Tukey boundaries are: **$\text{Lower Bound} = -2,257$, $\text{Upper Bound} = 4,007$**

Execution of the production script produced the following results:

- **Original rows:** 678,013
- **Density lower bound:** -2,257
- **Density upper bound:** 4,007
- **Cleaned rows:** 600,447
- **Rows removed:** 77,566

The cleaned dataset was successfully written to: `data/cleaned_business_data.csv`

The production pipeline is intentionally kept separate from exploratory business interpretation. During EDA, statistically unusual observations are evaluated in business context before being considered data-quality errors. The production script, however, consistently implements the required Tukey $1.5 \times \text{IQR}$ procedure to create a reproducible cleaned dataset while preserving the original raw dataset.

5.2 Reproducibility and GitHub Delivery

The project repository contains the required source code, notebook, exported data dictionary, generated figures, raw and cleaned datasets, dependency file, README instructions, and final PDF report.

The README.md provides step-by-step instructions to:

1. Clone the repository.
2. Create and activate a Python virtual environment.
3. Install dependencies from requirements.txt.
4. Execute the data-ingestion script.
5. Run the Jupyter Notebook.
6. Execute the production outlier-cleaning script.
7. Review the generated figures, data dictionary, cleaned dataset, and final report.

The final written report is compiled as: Module1 Homework Report.pdf

The complete repository will be pushed to the public GitHub repository before the repository URL is submitted through the Canvas Website URL submission field.

Conclusion

This project established a reproducible business analytics workflow for the French Motor Claims dataset by combining programmatic data procurement, structural data auditing, business interpretation, data-dictionary construction, visualization, correlation analysis, and production outlier management.

The analysis demonstrates an important distinction between a statistical outlier and a data-quality error. High values in variables such as Density, BonusMalus, ClaimNb, and VehAge may represent legitimate insurance behavior and therefore require business interpretation rather than automatic deletion during exploratory analysis.

ClaimNb is retained in its original count form because it represents discrete claim events and contains a large concentration of zero values. A direct logarithmic transformation is

inappropriate because $\log(0) = -\infty$. For claim-frequency modeling, ClaimNb can instead be modeled using a Poisson rate framework with $\log(\text{Exposure})$ as an offset.

Density is strongly right-skewed and may benefit from a $\log(1 + \text{Density})$ transformation for linear modeling. BonusMalus is retained as a continuous risk feature because high values may contain meaningful information, while VehAge can be retained and transformed or grouped depending on the modeling objective.

At the same time, the standalone production pipeline provides a reproducible implementation of the required Tukey $1.5 \times \text{IQR}$ filtering procedure. Together, the notebook, scripts, data dictionary, figures, cleaned dataset, README instructions, and final PDF provide a complete and reproducible analytics workflow.